Larvis (bioJARVIS): A Universal, State-Aware Wet-Lab AI Copilot

1. Problem, Users & Real-World Impact

Irreproducible preclinical research costs the US ~$28B annually (Freedman et al., 2015). A key bottleneck is the **gap between experimental execution and documentation**: researchers cannot continuously record observations while performing delicate procedures with gloved hands. Unnoticed bubbles, pipetting errors or temperature deviations can invalidate experiments and downstream results. **Larvis helps wet-lab researchers and technicians** monitor experiments, **identify deviations** and assess their **consequences before further resources are wasted.** Its modular architecture supports microfluidics, immunoassays, PCR, cell culture and chromatography. When errors occur, Larvis uses experimental data, protocol constraints and Bayesian optimisation to suggest corrective actions, helping researchers adapt their workflows while keeping final decisions in human hands.

2. Novelty & Multi-Step Agent Design

Unlike a conventional Gemini chatbot, Larvis integrates five specialised capabilities:

- **Gemini Live:** Bidirectional voice/vision interaction for hands-free guidance.
- **5-Agent Vision Council:** Segmentation, 3D geometry, temporal tracking, artefact rejection and consensus for physical anomaly detection.
- **Deterministic Steering:** Seven auditable actions check protocol constraints, trace downstream dependencies and investigate discrepancies.
- **Predictive Recovery:** PyTorch Yield MLP and Gaussian Process Bayesian Optimisation propose corrective adjustments.
- **Human-Controlled Audit:** Interactive dashboard, explicit approval gates and append-only SQLite records preserve experimental traceability.

Live proof of concept: A 12-channel microfluidic CEA cancer-biomarker immunoassay demonstrates anomaly detection, dependency tracking and recovery suggestions. Judges can submit novel voice/visual inputs.

3. Data, Scientific Grounding & Evaluation

Data: Five assay profiles, a 9-step protocol graph (33 dependencies), 18 synthetic optical frames and external evidence from PubMed, UniProt and PubChem. Protocol constraints reference ISO/CLSI guidance. Vision components include kLarity (AGPL-3.0/CC-BY 4.0) and BubbleID (MIT).

Additional results: 14/14 independent steering checks; Yield MLP R^2=0.979.

Evaluation	Stock Gemini	Larvis
Protocol accuracy (20 cases)	70%	**100%**
Bubble detection MAE (18 frames)	2.41	**0.12; F1 0.92**
Glare false-alarm rate	100%	**0%**
Blank-frame abstention	0%	**100%**
Voice routing	—	**15/15**

4. Agent Limits, Security & Safeguards

Limits: Evaluated using synthetic imagery and mock hardware; live-chip and cross-protocol reliability remain unvalidated. Pixel-based geometry requires calibration, and some inputs depend on spoken reports.

Safeguards: Independent vision verification, evidence-linked claims and abstention limit unsupported outputs. Human Accept/Dispute/Reject approval gates consequential changes. Append-only audit records, default-off sensors, ephemeral frames and short-lived API tokens protect data integrity and privacy.

Next step: Validate on real wet-lab footage with a UCL partner lab and expand testing across assay types.